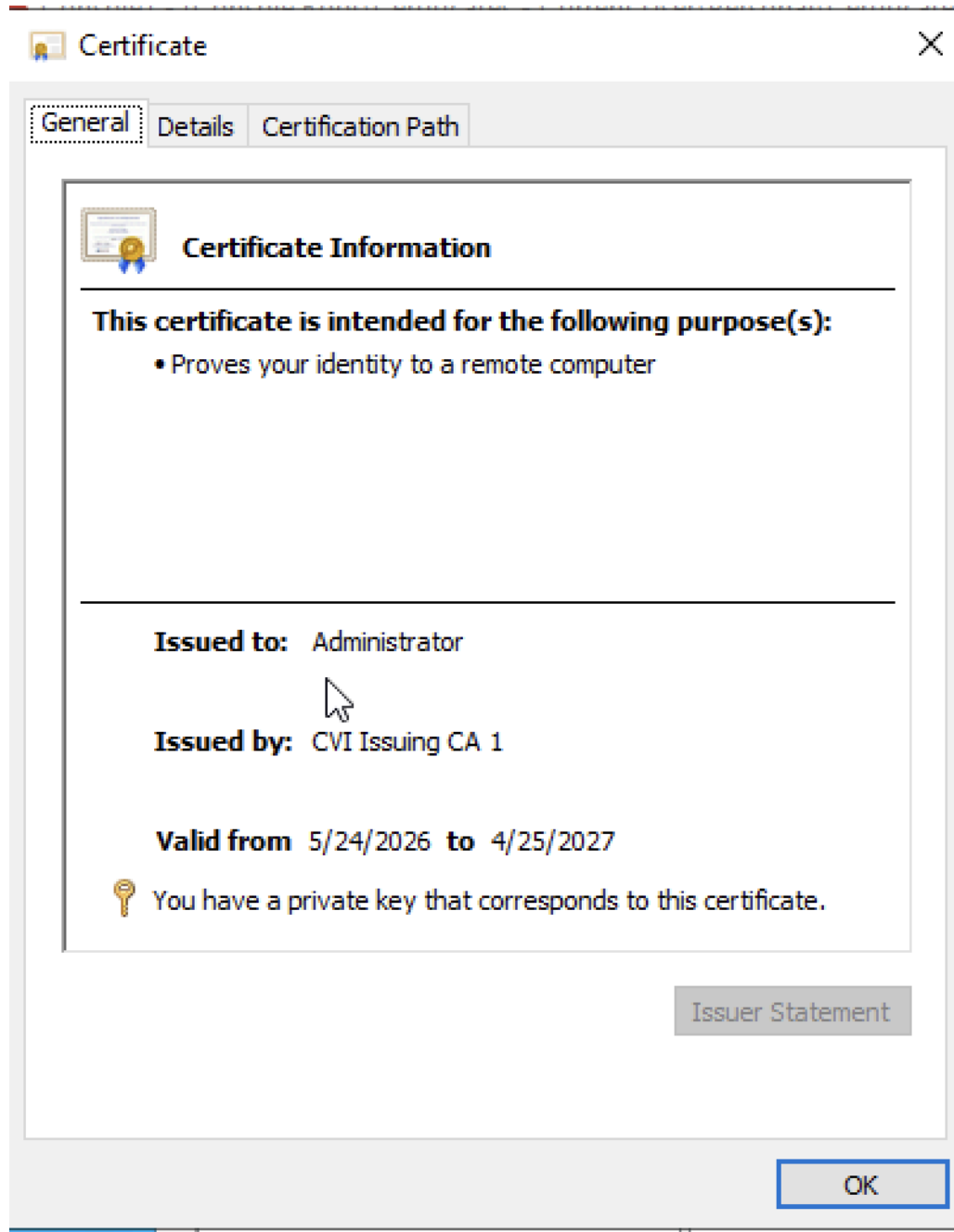
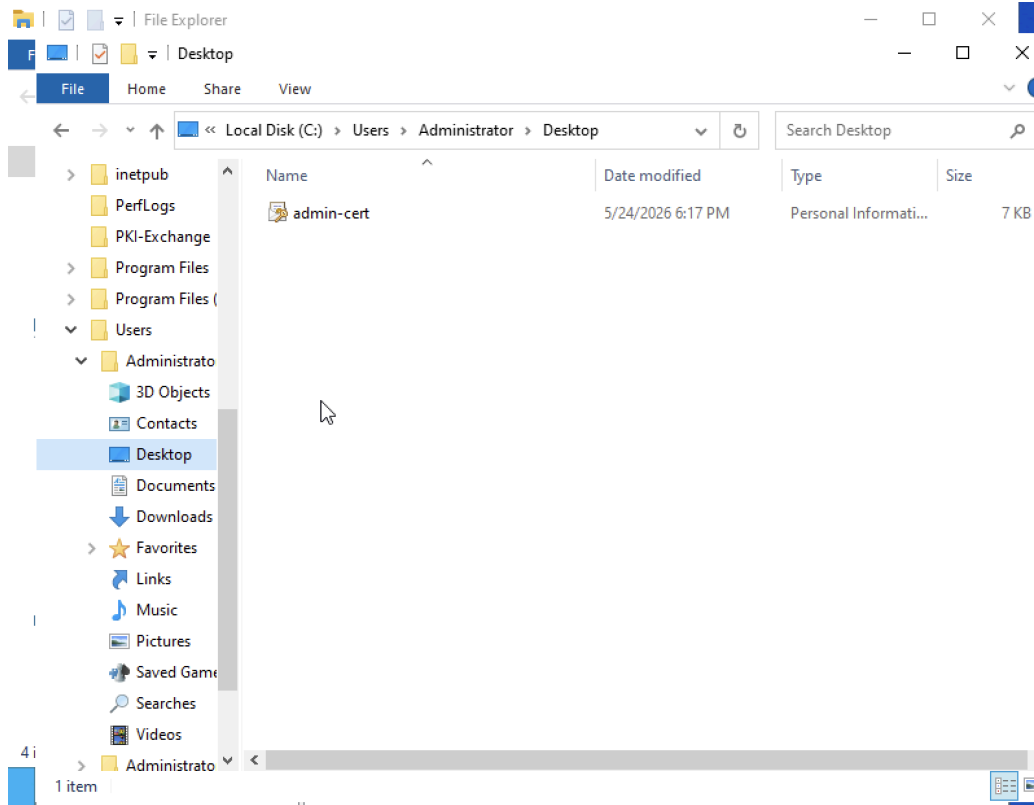


Omoze Idehenre
5/24/2026
PKI Week 11
Assets

Lab 1







```
Administrator: Windows PowerShell

Issuer: CN=CVI Issuing CA 1, DC=corp, DC=cvilab, DC=local
NotBefore: 5/17/2026 7:23 AM
NotAfter: 4/25/2027 7:36 PM
Subject: CN=cviissuingca1.corp.cvilib.local
Non-root Certificate
Template: CVI-WebServer
Cert Hash(sha1): 697926d7522ad1d1d3b1e0e5b271579e37fa45f4
  Key Container = 0ff881e482ec5b0a91e962d3f817606e_f0a99c17-76d3-498a-97de-2992c06105fd
  Simple container name: te-CVI-WebServer-6de1a47b-c00a-4c85-9a28-6539c8cccb8a
  Provider = Microsoft RSA SChannel Cryptographic Provider
Private key is NOT exportable
Encryption test passed

===== Certificate 1 =====
Serial Number: 580000002f7714edc7f317c4600000000002
Issuer: CN=CVI Root CA, DC=corp, DC=cvilib, DC=local
NotBefore: 4/25/2026 7:26 PM
NotAfter: 4/25/2027 7:36 PM
Subject: CN=CVI Issuing CA 1, DC=corp, DC=cvilib, DC=local
CA Version: V0.0
Certificate Template Name (Certificate Type): SubCA
Non-root Certificate
Template: SubCA, Subordinate Certification Authority
Cert Hash(sha1): 5137a597de2c3085ec5816c7f11edc18cfcdabaf8
  Key Container = CVI Issuing CA 1
  Unique container name: b52f658bb3f263e6f529f3a0187c63bc_f0a99c17-76d3-498a-97de-2992c06105fd
  Provider = Microsoft Software Key Storage Provider
Signature test passed
CertUtil: -store command completed successfully.
PS C:\Users\Administrator.CORP>
```

Lab 2

```
Administrator: Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator.CORP> Get-Service -Name CertSvc

Status      Name      DisplayName
-----
Running     CertSvc   Active Directory Certificate Services

PS C:\Users\Administrator.CORP> certutil -ping
Connecting to PKI-SRV01.corp.cvilab.local\CVI Issuing CA 1 ...
Server "CVI Issuing CA 1" ICertRequest2 interface is alive (94ms)
CertUtil: -ping command completed successfully.
PS C:\Users\Administrator.CORP> _
```

```
Administrator: Windows PowerShell
Issuer: CN=CVI Issuing CA 1, DC=corp, DC=cvilab, DC=local
NotBefore: 5/17/2026 7:23 AM
NotAfter: 4/25/2027 7:36 PM
Subject: CN=cviissuingca1.corp.cvilab.local
Non-root Certificate
Template: CVI-WebServer
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Key Container = 0ff881e482ec5b0a91e962d3f817606e_f0a99c17-76d3-498a-97de-2992c06105fd
Simple container name: te-CVI-WebServer-6de1a47b-c00a-4c85-9a28-6539c8cccb8a
Provider = Microsoft RSA Channel Cryptographic Provider
Private key is NOT exportable
Encryption test passed

===== Certificate 1 =====
Serial Number: 580000002f7714edc7f317c4600000000002
Issuer: CN=CVI Root CA, DC=corp, DC=cvilab, DC=local
NotBefore: 4/25/2026 7:26 PM
NotAfter: 4/25/2027 7:36 PM
Subject: CN=CVI Issuing CA 1, DC=corp, DC=cvilab, DC=local
CA Version: V0.0
Certificate Template Name (Certificate Type): SubCA
Non-root Certificate
Template: SubCA, Subordinate Certification Authority
Cert Hash(sha1): 5137a597de2c3085ec5816c7f11edc18cfcdbaf8
Key Container = CVI Issuing CA 1
Unique container name: b52f658bb3f263e6f529f3a0187c63bc_f0a99c17-76d3-498a-97de-2992c06105fd
Provider = Microsoft Software Key Storage Provider
Signature test passed
CertUtil: -store command completed successfully.
PS C:\Users\Administrator.CORP> _
```

Administrator: Windows PowerShell

```
Issuer: CN=CVI Issuing CA 1, DC=corp, DC=cvilab, DC=local
NotBefore: 5/17/2026 7:23 AM
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  Simple container name: te-CVI-WebServer-6de1a47b-c00a-4c85-9a28-6539c8cccb8a
  Provider = Microsoft RSA Channel Cryptographic Provider
Private key is NOT exportable
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===== Certificate 1 =====
Serial Number: 580000002f7714edc7f317c4600000000002
Issuer: CN=CVI Root CA, DC=corp, DC=cvilab, DC=local
NotBefore: 4/25/2026 7:26 PM
NotAfter: 4/25/2027 7:36 PM
Subject: CN=CVI Issuing CA 1, DC=corp, DC=cvilab, DC=local
CA Version: V0.0
Certificate Template Name (Certificate Type): SubCA
Non-root Certificate
Template: SubCA, Subordinate Certification Authority
Cert Hash(sha1): 5137a597de2c3085ec5816c7f11edc18cfcdabaf8
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  Provider = Microsoft Software Key Storage Provider
Signature test passed
CertUtil: -store command completed successfully.
PS C:\Users\Administrator.CORP>
```

pk1-serv01

Recycle Bin

Administrator: Windows PowerShell

Mode	LastWriteTime	Length	Name
-a----	5/24/2026 7:32 PM	184	Test-CVI.ps1

```
PS C:\Users\Administrator.CORP> $cert = Get-ChildItem Cert:\CurrentUser\My | Where-Object {$_.EnhancedKeyUsageList.FriendlyName -contains "Code Signing"} | Select-Object -First 1
PS C:\Users\Administrator.CORP> Set-AuthenticodeSignature "C:\Scripts\Test-CVI.ps1" $cert

Directory: C:\Scripts

SignerCertificate                                Status                                Path
-----
BD621E4E309E76534BA4D4A2EEE787268177C179 Valid                                Test-CVI.ps1

PS C:\Users\Administrator.CORP> Get-AuthenticodeSignature "C:\Scripts\Test-CVI.ps1"

Directory: C:\Scripts

SignerCertificate                                Status                                Path
-----
BD621E4E309E76534BA4D4A2EEE787268177C179 Valid                                Test-CVI.ps1

PS C:\Users\Administrator.CORP>
```

Lab 3

Administrator: Windows PowerShell

```
Issuer: CN=CVI Issuing CA 1, DC=corp, DC=cvilab, DC=local
NotBefore: 5/17/2026 7:23 AM
NotAfter: 4/25/2027 7:36 PM
Subject: CN=cviissuingca1.corp.cvilib.local
Non-root Certificate
Template: CVI-WebServer
Cert Hash(sha1): 697926d7522ad1d1d3b1e0e5b271579e37fa45f4
  Key Container = 0ff881e482ec5b0a91e962d3f817606e_f0a99c17-76d3-498a-97de-2992c06105fd
  Simple container name: te-CVI-WebServer-6de1a47b-c00a-4c85-9a28-6539c8cccb8a
  Provider = Microsoft RSA SChannel Cryptographic Provider
Private key is NOT exportable
Encryption test passed

===== Certificate 1 =====
Serial Number: 580000002f7714edc7f317c4600000000002
Issuer: CN=CVI Root CA, DC=corp, DC=cvilib, DC=local
NotBefore: 4/25/2026 7:26 PM
NotAfter: 4/25/2027 7:36 PM
Subject: CN=CVI Issuing CA 1, DC=corp, DC=cvilib, DC=local
CA Version: V0.0
Certificate Template Name (Certificate Type): SubCA
Non-root Certificate
Template: SubCA, Subordinate Certification Authority
Cert Hash(sha1): 5137a597de2c3085ec5816c7f11edc18cfcdabaf8
  Key Container = CVI Issuing CA 1
  Unique container name: b52f658bb3f263e6f529f3a0187c63bc_f0a99c17-76d3-498a-97de-2992c06105fd
  Provider = Microsoft Software Key Storage Provider
Signature test passed
CertUtil: -store command completed successfully.
PS C:\Users\Administrator.CORP>
```

Administrator: Windows PowerShell

```
Subject: CN=CVI Issuing CA 1, DC=corp, DC=cvilib, DC=local
CA Version: V0.0
Certificate Template Name (Certificate Type): SubCA
Non-root Certificate
Template: SubCA, Subordinate Certification Authority
Cert Hash(sha1): 5137a597de2c3085ec5816c7f11edc18cfcdabaf8
  Key Container = CVI Issuing CA 1
  Unique container name: b52f658bb3f263e6f529f3a0187c63bc_f0a99c17-76d3-498a-97de-2992c06105fd
  Provider = Microsoft Software Key Storage Provider
Signature test passed
CertUtil: -store command completed successfully.
PS C:\Users\Administrator.CORP> Get-ChildItem Cert:\CurrentUser\My | Format-List Subject, Thumbprint, EnhancedKeyUsageList

Subject           : CN=Administrator, CN=Users, DC=corp, DC=cvilib, DC=local
Thumbprint        : BD621E4E309E76534BA4D4A2EEE787268177C179
EnhancedKeyUsageList : {Code Signing (1.3.6.1.5.5.7.3.3)}

Subject           : CN=Administrator, CN=Users, DC=corp, DC=cvilib, DC=local
Thumbprint        : 0F39D2B15181C1B7511D222A19B930B2A287B38C
EnhancedKeyUsageList : {Client Authentication (1.3.6.1.5.5.7.3.2)}

Subject           : CN=Administrator, CN=Users, DC=corp, DC=cvilib, DC=local
Thumbprint        : 078DF6A9FB990103E4F59EE9F0E736DF404884CD
EnhancedKeyUsageList : {Client Authentication (1.3.6.1.5.5.7.3.2)}

PS C:\Users\Administrator.CORP>
```

```

PS C:\Users\Administrator.CORP> certutil -store My "697926D7522AD1D3B1E0E5B271579E37FA45F4
>> certutil -store My "0F39D2B15181C1B7511D222A19B930B2A287B8C
My "Personal"
CertUtil: -store command FAILED: 0x80090011 (-2146893807 NTE_NOT_FOUND)
CertUtil: Object was not found.
PS C:\Users\Administrator.CORP> certutil -store My 697926D7522AD1D3B1E0E5B271579E37FA45F4
My "Personal"
===== Certificate 0 =====
Serial Number: 440000003bf8819a444703a7500000000003
Issuer: CN=CVI Issuing CA 1, DC=corp, DC=cvilab, DC=local
NotBefore: 5/17/2026 7:23 AM
NotAfter: 4/25/2027 7:36 PM
Subject: CN=cviissuingca1.corp.cvilib.local
Non-root Certificate
Template: CVI-WebServer
Cert Hash(sha1): 697926d7522ad1d3b1e0e5b271579e37fa45f4
Key Container = 0ff881e482ec5b0a91e962d3f817606e_f0a99c17-76d3-498a-97de-2992c06105fd
Simple container name: te-CVI-WebServer-6de1a47b-c00a-4c85-9a28-6539c8cccb8a
Provider = Microsoft RSA SChannel Cryptographic Provider
Private key is NOT exportable
Encryption test passed
CertUtil: -store command completed successfully.
PS C:\Users\Administrator.CORP>

```

```

Administrator: Windows PowerShell
CertUtil: -store command FAILED: 0x80090011 (-2146893807 NTE_NOT_FOUND)
CertUtil: Object was not found.
PS C:\Users\Administrator.CORP> Get-ChildItem Cert:\CurrentUser\My | Format-List Subject,Issuer,Thumbprint,NotBefore,SerialNumber,EnhancedKeyUsageList

Subject          : CN=Administrator, CN=Users, DC=corp, DC=cvilab, DC=local
Issuer           : CN=CVI Issuing CA 1, DC=corp, DC=cvilab, DC=local
Thumbprint       : BD621E4E309E76534BA4D4A2EEE787268177C179
NotBefore        : 5/24/2026 7:01:32 PM
SerialNumber     : 4400000008DF33D73E94D83C56000000000008
EnhancedKeyUsageList : {Code Signing (1.3.6.1.5.5.7.3.3)}

Subject          : CN=Administrator, CN=Users, DC=corp, DC=cvilab, DC=local
Issuer           : CN=CVI Issuing CA 1, DC=corp, DC=cvilab, DC=local
Thumbprint       : 0F39D2B15181C1B7511D222A19B930B2A287B38C
NotBefore        : 5/24/2026 5:52:44 AM
SerialNumber     : 4400000006165D0A6376A728AE000000000006
EnhancedKeyUsageList : {Client Authentication (1.3.6.1.5.5.7.3.2)}

Subject          : CN=Administrator, CN=Users, DC=corp, DC=cvilab, DC=local
Issuer           : CN=CVI Issuing CA 1, DC=corp, DC=cvilab, DC=local
Thumbprint       : 078DF6A9FB990103E4F59EE9F0E736DF404B84CD
NotBefore        : 5/24/2026 5:53:24 PM
SerialNumber     : 440000000726EEFAAB089774B4000000000007
EnhancedKeyUsageList : {Client Authentication (1.3.6.1.5.5.7.3.2)}

PS C:\Users\Administrator.CORP>

```